

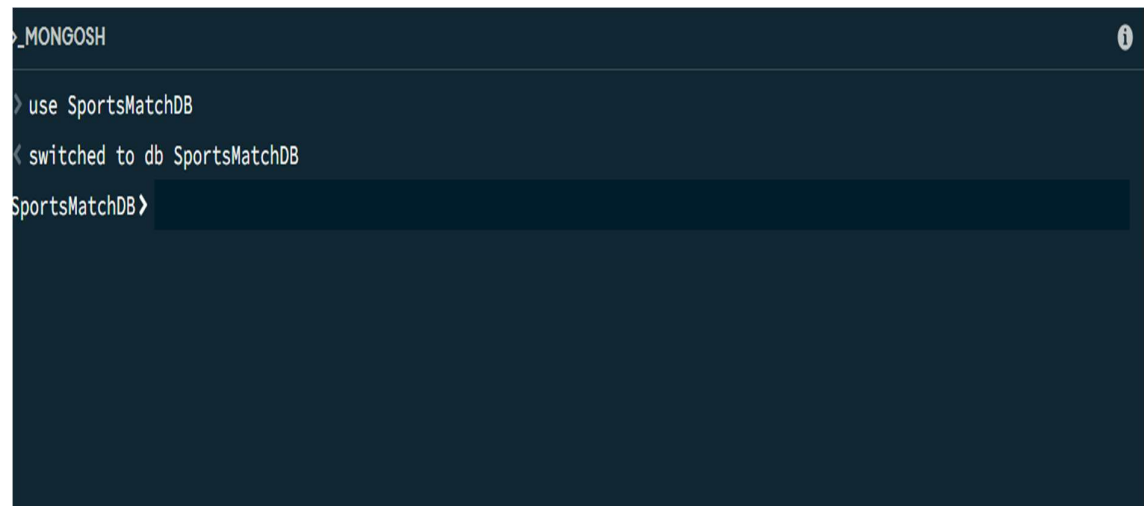
1. CREATE DATABASE

MongoDB Query:

use SportsMatchDB

Explanation: Creates or switches to the SportsMatchDB database.

Output Screenshot:



```
_MONGOSH
> use SportsMatchDB
< switched to db SportsMatchDB
SportsMatchDB>
```

2. CREATE COLLECTION

MongoDB Query:

db.createCollection("matches")

Explanation: Creates a collection named matches.

Output Screenshot:

```
> db.createCollection("matches")
< { ok: 1 }
SportsMatchDB > |
```

3. INSERT 60 FOOTBALL & HOCKEY MATCH RECORDS

MongoDB Query:

```
db.matches.insertMany([
  {matchId:1, sport:"Football", team1:"India", team2:"Japan", team1Score:3,
    team2Score:1, venue:"New Delhi"},
  {matchId:2, sport:"Football", team1:"Brazil", team2:"Argentina", team1Score:2,
    team2Score:2, venue:"Rio de Janeiro"},
  {matchId:3, sport:"Football", team1:"Germany", team2:"France", team1Score:1,
    team2Score:2, venue:"Berlin"},
  {matchId:4, sport:"Football", team1:"Spain", team2:"Italy", team1Score:3,
    team2Score:0, venue:"Madrid"},
  {matchId:5, sport:"Football", team1:"England", team2:"Portugal", team1Score:4,
    team2Score:2, venue:"London"},
  {matchId:6, sport:"Football", team1:"Netherlands", team2:"Belgium", team1Score:2,
    team2Score:1, venue:"Amsterdam"},
  {matchId:7, sport:"Football", team1:"Argentina", team2:"Uruguay", team1Score:3,
    team2Score:1, venue:"Buenos Aires"},
  {matchId:8, sport:"Football", team1:"Brazil", team2:"Colombia", team1Score:2,
    team2Score:0, venue:"Sao Paulo"},
  {matchId:9, sport:"Football", team1:"France", team2:"Belgium", team1Score:1,
    team2Score:1, venue:"Paris"},
  {matchId:10, sport:"Football", team1:"Italy", team2:"Croatia", team1Score:0,
    team2Score:1, venue:"Rome"},

  {matchId:11, sport:"Football", team1:"India", team2:"Australia", team1Score:2,
    team2Score:1, venue:"Mumbai"},
  {matchId:12, sport:"Football", team1:"Japan", team2:"South Korea", team1Score:1,
    team2Score:1, venue:"Tokyo"},
  {matchId:13, sport:"Football", team1:"USA", team2:"Mexico", team1Score:3,
    team2Score:2, venue:"Dallas"},
  {matchId:14, sport:"Football", team1:"Canada", team2:"USA", team1Score:0,
    team2Score:2, venue:"Toronto"},
  {matchId:15, sport:"Football", team1:"Portugal", team2:"Spain", team1Score:2,
    team2Score:3, venue:"Lisbon"},
  {matchId:16, sport:"Football", team1:"Germany", team2:"Netherlands", team1Score:4,
    team2Score:1, venue:"Munich"},
  {matchId:17, sport:"Football", team1:"England", team2:"Germany", team1Score:2,
    team2Score:0, venue:"Manchester"},
  {matchId:18, sport:"Football", team1:"France", team2:"Spain", team1Score:3,
    team2Score:2, venue:"Lyon"},
  {matchId:19, sport:"Football", team1:"Brazil", team2:"Chile", team1Score:1,
    team2Score:0, venue:"Brasilia"},
```

{matchId:20, sport:"Football", team1:"Argentina", team2:"Brazil", team1Score:2, team2Score:1, venue:"Cordoba"},

{matchId:21, sport:"Football", team1:"India", team2:"Thailand", team1Score:4, team2Score:0, venue:"Kolkata"},

{matchId:22, sport:"Football", team1:"Australia", team2:"Japan", team1Score:2, team2Score:2, venue:"Sydney"},

{matchId:23, sport:"Football", team1:"Spain", team2:"Germany", team1Score:1, team2Score:1, venue:"Barcelona"},

{matchId:24, sport:"Football", team1:"Italy", team2:"France", team1Score:0, team2Score:3, venue:"Milan"},

{matchId:25, sport:"Football", team1:"Portugal", team2:"Belgium", team1Score:3, team2Score:1, venue:"Porto"},

{matchId:26, sport:"Football", team1:"England", team2:"Netherlands", team1Score:2, team2Score:2, venue:"Liverpool"},

{matchId:27, sport:"Football", team1:"USA", team2:"Canada", team1Score:3, team2Score:1, venue:"Boston"},

{matchId:28, sport:"Football", team1:"Mexico", team2:"Costa Rica", team1Score:2, team2Score:0, venue:"Mexico City"},

{matchId:29, sport:"Football", team1:"South Korea", team2:"China", team1Score:1, team2Score:0, venue:"Seoul"},

{matchId:30, sport:"Football", team1:"Japan", team2:"Iran", team1Score:2, team2Score:1, venue:"Osaka"},

{matchId:31, sport:"Hockey", team1:"India", team2:"Australia", team1Score:4, team2Score:3, venue:"Bhubaneswar"},

{matchId:32, sport:"Hockey", team1:"Germany", team2:"Netherlands", team1Score:2, team2Score:3, venue:"Berlin"},

{matchId:33, sport:"Hockey", team1:"England", team2:"Belgium", team1Score:3, team2Score:2, venue:"London"},

{matchId:34, sport:"Hockey", team1:"Argentina", team2:"Spain", team1Score:4, team2Score:1, venue:"Buenos Aires"},

{matchId:35, sport:"Hockey", team1:"India", team2:"Germany", team1Score:5, team2Score:2, venue:"Rourkela"},

{matchId:36, sport:"Hockey", team1:"Australia", team2:"Belgium", team1Score:3, team2Score:3, venue:"Sydney"},

{matchId:37, sport:"Hockey", team1:"Netherlands", team2:"England", team1Score:4, team2Score:2, venue:"Amsterdam"},

{matchId:38, sport:"Hockey", team1:"Germany", team2:"Argentina", team1Score:2, team2Score:4, venue:"Hamburg"},

{matchId:39, sport:"Hockey", team1:"India", team2:"Netherlands", team1Score:3, team2Score:2, venue:"New Delhi"},

{matchId:40, sport:"Hockey", team1:"Belgium", team2:"Spain", team1Score:5, team2Score:1, venue:"Brussels"},

```

{matchId:41, sport:"Hockey", team1:"Australia", team2:"India", team1Score:2,
team2Score:3, venue:"Melbourne"},
{matchId:42, sport:"Hockey", team1:"England", team2:"Germany", team1Score:1,
team2Score:4, venue:"Manchester"},
{matchId:43, sport:"Hockey", team1:"Argentina", team2:"Netherlands", team1Score:3,
team2Score:3, venue:"Cordoba"},
{matchId:44, sport:"Hockey", team1:"Belgium", team2:"India", team1Score:2,
team2Score:4, venue:"Antwerp"},
{matchId:45, sport:"Hockey", team1:"Spain", team2:"England", team1Score:3,
team2Score:1, venue:"Barcelona"},
{matchId:46, sport:"Hockey", team1:"Germany", team2:"Australia", team1Score:4,
team2Score:2, venue:"Munich"},
{matchId:47, sport:"Hockey", team1:"Netherlands", team2:"Belgium", team1Score:3,
team2Score:2, venue:"Rotterdam"},
{matchId:48, sport:"Hockey", team1:"India", team2:"Argentina", team1Score:2,
team2Score:2, venue:"Mumbai"},
{matchId:49, sport:"Hockey", team1:"Australia", team2:"England", team1Score:5,
team2Score:3, venue:"Perth"},
{matchId:50, sport:"Hockey", team1:"Germany", team2:"Spain", team1Score:3,
team2Score:1, venue:"Frankfurt"},

{matchId:51, sport:"Hockey", team1:"India", team2:"Belgium", team1Score:4,
team2Score:2, venue:"Chennai"},
{matchId:52, sport:"Hockey", team1:"Netherlands", team2:"Australia", team1Score:2,
team2Score:2, venue:"The Hague"},
{matchId:53, sport:"Hockey", team1:"England", team2:"Argentina", team1Score:3,
team2Score:4, venue:"London"},
{matchId:54, sport:"Hockey", team1:"Spain", team2:"India", team1Score:1,
team2Score:3, venue:"Madrid"},
{matchId:55, sport:"Hockey", team1:"Belgium", team2:"Germany", team1Score:2,
team2Score:3, venue:"Brussels"},
{matchId:56, sport:"Hockey", team1:"Australia", team2:"Argentina", team1Score:4,
team2Score:2, venue:"Canberra"},
{matchId:57, sport:"Hockey", team1:"India", team2:"England", team1Score:3,
team2Score:1, venue:"Bengaluru"},
{matchId:58, sport:"Hockey", team1:"Netherlands", team2:"Germany", team1Score:2,
team2Score:1, venue:"Utrecht"},
{matchId:59, sport:"Hockey", team1:"Argentina", team2:"Belgium", team1Score:4,
team2Score:3, venue:"Rosario"},
{matchId:60, sport:"Hockey", team1:"Spain", team2:"Australia", team1Score:2,
team2Score:4, venue:"Valencia"}

```

])

Explanation: Inserts exactly **60 match documents** — **30 Football** and **30 Hockey** matches.

Output Screenshot:

```
  ])  
< {  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('6abfc062490b70eefaed0504'),  
    '1': ObjectId('6abfc062490b70eefaed0505'),  
    '2': ObjectId('6abfc062490b70eefaed0506'),  
    '3': ObjectId('6abfc062490b70eefaed0507'),  
    '4': ObjectId('6abfc062490b70eefaed0508'),  
    '5': ObjectId('6abfc062490b70eefaed0509'),  
    '6': ObjectId('6abfc062490b70eefaed050a'),  
    '7': ObjectId('6abfc062490b70eefaed050b'),  
    '8': ObjectId('6abfc062490b70eefaed050c'),  
    '9': ObjectId('6abfc062490b70eefaed050d'),  
    '10': ObjectId('6abfc062490b70eefaed050e'),  
    '11': ObjectId('6abfc062490b70eefaed050f'),  
    '12': ObjectId('6abfc062490b70eefaed0510'),  
    '13': ObjectId('6abfc062490b70eefaed0511'),  
    '14': ObjectId('6abfc062490b70eefaed0512'),  
    '15': ObjectId('6abfc062490b70eefaed0513'),  
    '16': ObjectId('6abfc062490b70eefaed0514'),
```

4. DISPLAY ALL MATCHES

MongoDB Query:

`db.matches.find()`

Explanation: Retrieves all match records from the matches collection.

Output Screenshots:

```
> db.matches.find()
< {
  _id: ObjectId('6abfc062490b70eefaed0504'),
  matchId: 1,
  sport: 'Football',
  team1: 'India',
  team2: 'Japan',
  team1Score: 3,
  team2Score: 1,
  venue: 'New Delhi'
}
{
  _id: ObjectId('6abfc062490b70eefaed0505'),
  matchId: 2,
  sport: 'Football',
  team1: 'Brazil',
  team2: 'Argentina',
  team1Score: 2,
  team2Score: 2,
  venue: 'Rio de Janeiro'
}
{
  _id: ObjectId('6abfc062490b70eefaed0506'),
  matchId: 3,
  sport: 'Football',
  team1: 'Germany',
  team2: 'France',
  team1Score: 1,
  team2Score: 2,
  venue: 'Berlin'
}
{
  _id: ObjectId('6abfc062490b70eefaed0507'),
  matchId: 4,
  sport: 'Football',
  team1: 'Spain',
  team2: 'Italy',
  team1Score: 3,
  team2Score: 0,
  venue: 'Madrid'
}
```

5. EQUAL TO — \$eq

Find matches where Team 1 scored exactly 3 goals.

MongoDB Query:

```
db.matches.find({
  team1Score: {$eq: 3}
})
```

Explanation: Displays matches where team1Score is exactly 3.

Output Screenshot:



```
> db.matches.find({
  team1Score: {$eq: 3}
})
< {
  _id: ObjectId('6abfc062490b70eefaed0504'),
  matchId: 1,
  sport: 'Football',
  team1: 'India',
  team2: 'Japan',
  team1Score: 3,
  team2Score: 1,
  venue: 'New Delhi'
}
{
  _id: ObjectId('6abfc062490b70eefaed0507'),
  matchId: 4,
  sport: 'Football',
  team1: 'Spain',
  team2: 'Italy',
  team1Score: 3,
  team2Score: 0,
  venue: 'Madrid'
}
```

6. GREATER THAN — \$gt

MongoDB Query:

```
db.matches.find({
```

```
    team1Score: {$gt: 3}
  })
```

Explanation: Finds matches where Team 1 scored more than 3 goals.

Output Screenshot:

```
> db.matches.find({
  team1Score: {$gt: 3}
})
< {
  _id: ObjectId('6abfc062490b70eefaed0508'),
  matchId: 5,
  sport: 'Football',
  team1: 'England',
  team2: 'Portugal',
  team1Score: 4,
  team2Score: 2,
  venue: 'London'
}
{
  _id: ObjectId('6abfc062490b70eefaed0513'),
  matchId: 16,
  sport: 'Football',
  team1: 'Germany',
  team2: 'Netherlands',
  team1Score: 4,
  team2Score: 1,
  venue: 'Munich'
}
```

7. LESS THAN — \$lt

MongoDB Query:

```
db.matches.find({
```

```
    team1Score: {$lt: 2}
  })
```

Explanation: Finds matches where Team 1 scored fewer than 2 goals.

Output Screenshot:

```
> db.matches.find({
  team1Score: {$gt: 3}
})
< {
  _id: ObjectId('6abfc062490b70eefaed0508'),
  matchId: 5,
  sport: 'Football',
  team1: 'England',
  team2: 'Portugal',
  team1Score: 4,
  team2Score: 2,
  venue: 'London'
}
{
  _id: ObjectId('6abfc062490b70eefaed0513'),
  matchId: 16,
  sport: 'Football',
  team1: 'Germany',
  team2: 'Netherlands',
  team1Score: 4,
  team2Score: 1,
  venue: 'Munich'
}
```

8. GREATER THAN OR EQUAL TO — \$gte

MongoDB Query:

```
db.matches.find({
```

```
team1Score: {$gte: 4}
})
```

Explanation: Finds matches where Team 1 scored 4 or more goals.

Output Screenshot:

```
> db.matches.find({
  team1Score: {$gte: 4}
})
< {
  _id: ObjectId('6abfc062490b70eefaed0508'),
  matchId: 5,
  sport: 'Football',
  team1: 'England',
  team2: 'Portugal',
  team1Score: 4,
  team2Score: 2,
  venue: 'London'
}
{
  _id: ObjectId('6abfc062490b70eefaed0513'),
  matchId: 16,
  sport: 'Football',
  team1: 'Germany',
  team2: 'Netherlands',
  team1Score: 4,
  team2Score: 1,
  venue: 'Munich'
}
```

9. LESS THAN OR EQUAL TO — \$lte

MongoDB Query:

```
db.matches.find({
```

```
team1Score: {$lte: 2}
})
```

Explanation: Finds matches where Team 1 scored 2 or fewer goals.

Output Screenshot:

```
> db.matches.find({
  team1Score: {$lte: 2}
})
< {
  _id: ObjectId('6abfc062490b70eefaed0505'),
  matchId: 2,
  sport: 'Football',
  team1: 'Brazil',
  team2: 'Argentina',
  team1Score: 2,
  team2Score: 2,
  venue: 'Rio de Janeiro'
}
{
  _id: ObjectId('6abfc062490b70eefaed0506'),
  matchId: 3,
  sport: 'Football',
  team1: 'Germany',
  team2: 'France',
  team1Score: 1,
  team2Score: 2,
  venue: 'Berlin'
}
```

10. UPDATE OPERATION

Update the score of match number 5.

MongoDB Query:

```
db.matches.updateOne(  
  {matchId:5},  
  {$set:{team1Score:5}}  
)
```

Explanation: Updates the Team 1 score of match 5 from 4 to 5.

Output Screenshot:



```
> db.matches.updateOne(  
  {matchId:5},  
  {$set:{team1Score:5}}  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}  
SportsMatchDB >  
> db.matches.find({matchId:5})  
< {  
  _id: ObjectId('6abfc062490b70eefaed0508'),  
  matchId: 5,  
  sport: 'Football',  
  team1: 'England',  
  team2: 'Portugal',  
  team1Score: 5,  
  team2Score: 2,  
  venue: 'London'  
}
```

11. DELETE ONE OPERATION

MongoDB Query:

```
db.matches.deleteOne({
```

matchId:10

}}

Explanation: Deletes the match having matchId equal to 10.

Output Screenshot:

```
> db.matches.deleteOne({
  matchId:10
})
< {
  acknowledged: true,
  deletedCount: 1
}
```

12. DELETE MANY OPERATION

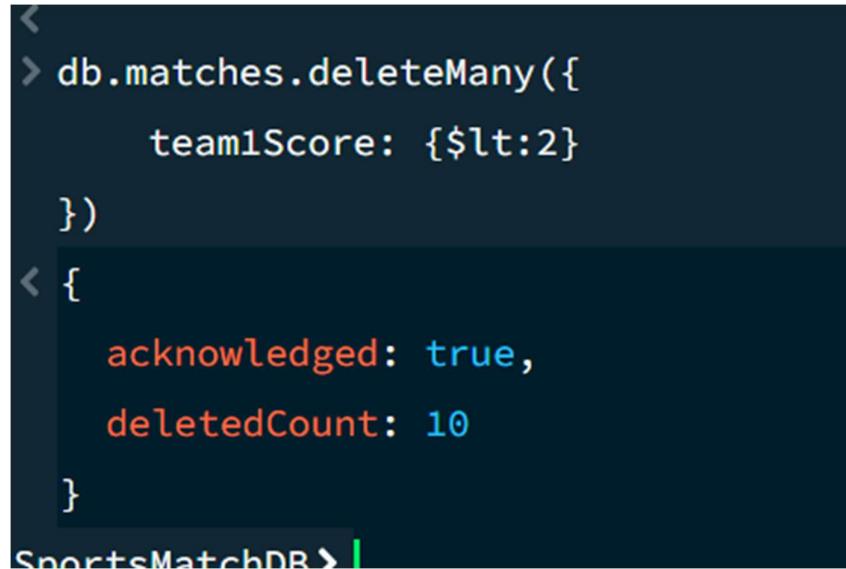
MongoDB Query:

db.matches.deleteMany({

```
team1Score: {$lt:2}
})
```

Explanation: Deletes all matches where Team 1 scored less than 2 goals.

Output Screenshot:



```
<
> db.matches.deleteMany({
    team1Score: {$lt:2}
})
< {
  acknowledged: true,
  deletedCount: 10
}
SportsMatchDB> |
```

13. AND OPERATION — \$and

Find matches where:

- Team 1 scored more than 3 goals
- Sport is Hockey

MongoDB Query:

```
db.matches.find({
  $and:[
    {team1Score:{$gt:3}},
    {sport:"Hockey"}
  ]
})
```

Explanation: Both conditions must be true.

Output Screenshot:

```
> db.matches.find({
  $and:[
    {team1Score:{$gt:3}},
    {sport:"Hockey"}
  ]
})
< {
  _id: ObjectId('6abfc062490b70eefaed0522'),
  matchId: 31,
  sport: 'Hockey',
  team1: 'India',
  team2: 'Australia',
  team1Score: 4,
  team2Score: 3,
  venue: 'Bhubaneswar'
}
{
  _id: ObjectId('6abfc062490b70eefaed0525'),
  matchId: 34,
  sport: 'Hockey',
  team1: 'Argentina',
  team2: 'Spain',
  team1Score: 4,
  team2Score: 1,
  venue: 'Buenos Aires'
}
```

14. OR OPERATION — \$or

Find matches involving either India or Australia as Team 1.

MongoDB Query:

```
db.matches.find({
  $or:[
    {team1:"India"},
    {team1:"Australia"}
  ]
})
```

Explanation: At least one of the conditions must be true.

Output Screenshot:



```
> db.matches.find({
  $or:[
    {team1:"India"},
    {team1:"Australia"}
  ]
})
< {
  _id: ObjectId('6abfc062490b70eefaed0504'),
  matchId: 1,
  sport: 'Football',
  team1: 'India',
  team2: 'Japan',
  team1Score: 3,
  team2Score: 1,
  venue: 'New Delhi'
}
{
  _id: ObjectId('6abfc062490b70eefaed050e'),
  matchId: 11,
  sport: 'Football',
  team1: 'India',
  team2: 'Australia',
  team1Score: 2,
  team2Score: 1,
  venue: 'Mumbai'
}
```

15. SORT OPERATION

A. Ascending Order

MongoDB Query:

```
db.matches.find().sort({  
  team1Score:1  
})
```

Explanation: Sorts matches from the lowest Team 1 score to the highest.

1 = ascending order.

Output Screenshot:



```
> db.matches.find().sort({  
  team1Score:1  
})  
< {  
  _id: ObjectId('6abfc062490b70eefaed050b'),  
  matchId: 8,  
  sport: 'Football',  
  team1: 'Brazil',  
  team2: 'Colombia',  
  team1Score: 2,  
  team2Score: 0,  
  venue: 'Sao Paulo'  
}  
{  
  _id: ObjectId('6abfc062490b70eefaed0505'),  
  matchId: 2,  
  sport: 'Football',  
  team1: 'Brazil',  
  team2: 'Argentina',  
  team1Score: 2,  
  team2Score: 2,  
  venue: 'Rio de Janeiro'  
}  
}
```

B. Descending Order

MongoDB Query:

```
db.matches.find().sort({
  team1Score:-1
})
```

Explanation: Sorts matches from the highest Team 1 score to the lowest.

-1 = descending order.

Output Screenshot:

```
> db.matches.find().sort({
  team1Score:-1
})
< {
  _id: ObjectId('6abfc062490b70eefaed0508'),
  matchId: 5,
  sport: 'Football',
  team1: 'England',
  team2: 'Portugal',
  team1Score: 5,
  team2Score: 2,
  venue: 'London'
}
{
  _id: ObjectId('6abfc062490b70eefaed0526'),
  matchId: 35,
  sport: 'Hockey',
  team1: 'India',
  team2: 'Germany',
  team1Score: 5,
  team2Score: 2,
  venue: 'Rourkela'
}
```

16. LIMIT OPERATION

Display only the first five matches:

MongoDB Query:

db.matches.find().limit(5)

Explanation: Displays only the first five documents.

Output Screenshot:

```
> db.matches.find().limit(5)
< {
  _id: ObjectId('6abfc062490b70eefaed0504'),
  matchId: 1,
  sport: 'Football',
  team1: 'India',
  team2: 'Japan',
  team1Score: 3,
  team2Score: 1,
  venue: 'New Delhi'
}
{
  _id: ObjectId('6abfc062490b70eefaed0505'),
  matchId: 2,
  sport: 'Football',
  team1: 'Brazil',
  team2: 'Argentina',
  team1Score: 2,
  team2Score: 2,
  venue: 'Rio de Janeiro'
}
```

17. SEARCH BY TEAM

Find all matches where India was Team 1:

MongoDB Query:

```
db.matches.find({
  team1:"India"
})
```

Explanation: Retrieves all matches in which India appears as team1.

Output Screenshot:



```
> db.matches.find({
  team1:"India"
})
< {
  _id: ObjectId('6abfc062490b70eefaed0504'),
  matchId: 1,
  sport: 'Football',
  team1: 'India',
  team2: 'Japan',
  team1Score: 3,
  team2Score: 1,
  venue: 'New Delhi'
}
{
  _id: ObjectId('6abfc062490b70eefaed050e'),
  matchId: 11,
  sport: 'Football',
  team1: 'India',
  team2: 'Australia',
  team1Score: 2,
  team2Score: 1,
  venue: 'Mumbai'
}
```

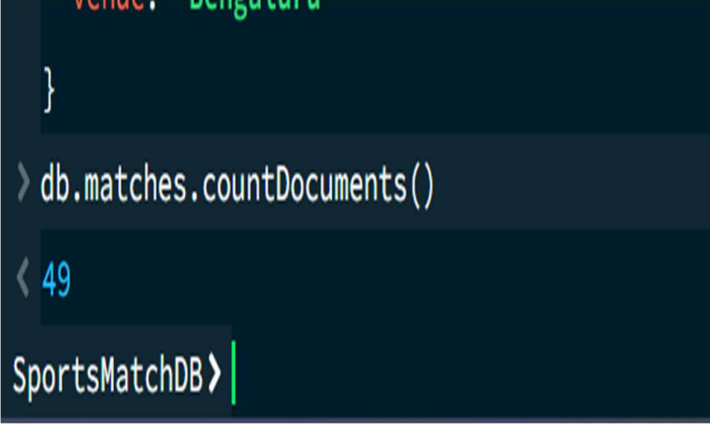
18. COUNT DOCUMENTS

MongoDB Query:

db.matches.countDocuments()

Explanation: Counts the documents currently present in the matches collection.

Output Screenshot:



```
    }  
  }  
  > db.matches.countDocuments()  
  < 49  
SportsMatchDB > |
```

